Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/25/2024
PI Name:	Iyer-Pascuzzi, Anjali Ph.D.		
Project Title:	Effect of Spaceflight and Simulated Microgravity on Plant Defense Responses		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2054	**Congressional District:**	4
Comments:			
Project Type:	Flight,Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	09/01/2019	**End Date:**	08/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2025 per NSSC information (Ed., 7/3/24). NOTE: End date changed to 08/31/2024 per NSSC information (Ed., 2/9/24). NOTE: End date changed to 08/31/2023 per NSSC information (Ed., 6/6/22).		
Key Personnel Changes/Previous PI:	June 2024 report: No changes to personnel. June 2023 report: No changes to personnel. June 2022 report: No changes to personnel. June 2021 report: No changes to personnel. June 2020 report: No changes to personnel.		
COI Name (Institution):	Sparks, Erin Ph.D. (University of Delaware)		
Grant/Contract No.:	80NSSC19K1495		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As on Earth, in space plants must defend themselves against microbial and fungal pathogens. For example, plants grown on the International Space Station (ISS) recently became diseased from the fungal pathogen Fusarium oxysporum. Although previous studies have demonstrated that plant defense pathways are altered by spaceflight when plants were grown in sterile conditions, fundamental knowledge of how the plant immune system responds to microbes or defense elicitors during spaceflight is lacking. This knowledge is critical to growing plants that can withstand the rigors of long duration spaceflight, and is particularly important in a horticultural crop that will provide nutrition to the space crew. In this proposal, we will perform the following objectives: 1) Investigate physiological and whole genome transcriptional responses to defense activation in wild-type and immune-deficient tomatoes during spaceflight. Tomatoes will be grown in the Advanced Plant Habitat (APH). We will activate defense responses with a chemical elicitor. At 24 and 48 hours after defense activation, we will harvest tissue and subsequently perform next-generation sequencing to identify genome-wide transcriptional defense responses. In addition, we will use next-generation sequencing to examine the transcriptional response to spaceflight in immune-deficient tomatoes. All plants will be imaged daily to understand the impact of spaceflight on growth rates of immune-activated and immune-deficient tomatoes. All experiments will be performed in parallel on the ground. 2) Determine whether colonization of tomato by the fungal plant pathogen Fusarium oxysporum is impacted by simulated microgravity. We will grow plants in a 2D-ground-based microgravity simulator and inoculate them with Fusarium oxysporum. We will assess tomato plant colonization using histological techniques. This work will generate key fundamental knowledge of plant-microbe interactions that is important for understanding plant production in space. It is consistent with the goal of the Plant Biology Element in the Space Biology Science Plan 2016-2025.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	On Earth, plant pathogens cause upwards of 20–40% global crop loss from disease. Microbes, including plant pathogens, have been isolated from the International Space Station (ISS), and thus disease loss also threatens crop production in space. Spaceflight studies are important to life on Earth because these studies have the potential to uncover new interactions between gravity and plant responses to pathogens. For example, the plant hormone auxin is known for its critical roles in plant development and defense. Recent experiments on the ISS revealed novel interactions between gravity and auxin. This new information will be informative for developing new plant disease resistance strategies on Earth and will enhance our ability to grow crops on Earth.
Task Progress:	Summary of research accomplishment 09/01/23 – 08/31/24 Specific Aim 1: Determine the effect of spaceflight on genome-wide transcriptomic and physiological defense responses in tomatoes. i) Performed EVT tests in the APH with Red Wire team and SPACEFLIGHT! Work in 2023-2043 was aimed at finalizing tests for flight. With the team at Red Wire, we completed a successful Experiment Verification Test (EVT) in the summer of 2023 and we launched tomato seeds to the International Space Station (ISS) on SpaceX-29 on Nov. 10, 2023! The science carrier was installed in the Advanced Plant Habitat (APH) and flood filled by Japanese Space Agency (JAXA) astronaut crewmember Satoshi on Dec. 30, 2023. Seeds germinated in late December 2023, and experiments were performed by astronauts on the ISS in January 2024. The experiment was executed according to the Flight Operations Flow Chart. The experiment went very well both in space and in the ground controls. There were no unexpected complications, other than the Sharpie that was used to mark leaves did not work as well in space as on the ground (it did work, but required much patience from the astronaut). We had excellent germination of both wild type Moneymaker plants and mutant NahG plants, plant thinning, swabbing of salicylic acid (SA), and leaf harvest. Three different astronauts performed the experiments, and we were able to watch and communicate with them throughout the experiment. All of our success criteria were excellent. Leaves were stowed in the Minus Eighty (Degrees Celsius) Laboratory Freezer for ISS (MELFI) (-80C) in the ISS and returned to Earth on AX-3 Dragon Capsule on Feb. 9, 2024. We received them at Purdue the following week. We ground tissue and extracted RNA from leaves of all flight and ground samples. One ground sample did not have sufficient tissue to generate RNA. Thus, we had 36 flight samples (100%) and 35 ground samples (97%). All (100%) of our samples were good quality RNA. In June 2024, the samples were sent to Novogene for sequencing. We anticipate receiving sequencing results in August and sharing the results at the European Low Gravity Research Association (ELGRA) meeting in Liverpool in September 2024 and at the American Society for Gravitational and Space Research (ASGSR) in Puerto Rico in December 2024. We also plan to measure plant growth in spaceflight using the downloaded images from the APH in spaceflight and in ground controls. Specific Aim 2: Investigate how simulated microgravity affects fungal colonization of tomato plants. In Aim 2, we proposed a series of ground-based experiments to investigate the impact of simulated microgravity on the ability of fungal pathogens to infect tomato roots. In the first year of this project, we redesigned the 2D clinostat to accommodate eight plants as opposed to the original four-plant design. Using our redesigned system, we showed we could grow tomato cultivars to maturity under continuous clinorotation. In years 2-4, we successfully demonstrated the infection of Moneymaker tomato plants by Fusarium oxysporum in the enclosed rhizoboxes. We also built a second clinostat to simultaneously analyze plants under clinorotation perpendicular and parallel to the gravity vector. We also optimized the growth conditions on the paired clinostat (clinorotated and upright) and established conditions in which the plants were healthy for 15 days on the clinostat. In 2023-2024, we have continued to grow and inoculate plants to increase the number of replicates. We are currently sectioning roots and investigating fungal growth. In addition to the microscopy work, in 2023-2024, the Sparks lab partnered with DeNovix, Inc for capacity enhancement of this project. DeNovix is an instrumentation company located in Wilmington, DE. Taking advantage of the Delaware Biosciences Center for Advance Technology (CAT) Technology Access Program (TAP), DeNovix received a $5000 grant with committed matching funds to explore the use of their CellDrop instrument for counting cell nuclei. For the

	experimental system, DeNovix asked if the Sparks lab would partner to provide samples, thus ensuring there was a scientific outcome of this system validation. The funding supported the analysis of 4 samples, and we selected two plants each of upright and clinorotated (H7996 cultivar) for analysis. Nuclei were isolated from root tips and then subjected to 10X Genomics capture and sequencing. These data are currently being analyzed.
Bibliography Type:	Description: (Last Updated: 04/15/2026)
Abstracts for Journals and Proceedings	Iyer-Pascuzzi AS. "Sustainable food production in space: Elucidating the impact of spaceflight on plant-microbe interactions. Keynote session." Plant Biology 2023 (American Society of Plant Biologists annual meeting), Savannah, Georgia, August 5-9, 2023. Abstracts. Plant Biology 2023 (American Society of Plant Biologists annual meeting), Savannah, Georgia, August 5-9, 2023. , Aug-2023
Abstracts for Journals and Proceedings	Iyer-Pascuzzi AS. "Sustainable food production in space: The challenge of plant-microbe interactions." Department of Crop Sciences, University of Illinois, Urbana, IL, April 2024. Report on invited talk. Department of Crop Sciences, University of Illinois, Urbana, IL, April 2024. , Apr-2024
Awards	Iyer-Pascuzzi AS. "Distinguished Researcher, and one of 25 Inspiring Women in Plant Biology from the American Society of Plant Biologists (ASPB), May 15, 2024." May-2024
Significant Media Coverage	Black S. (Iyer-Pascuzzi A feature). "25 inspiring women in plant biology. Dr. Iyer-Pascuzzi featured in blog post on distinguished researchers." American Society of Plant Biologists (ASPB) blog, May 15, 2024, https://blog.aspb.org/25-inspiring-women-in-plant-biology/ , May-2024